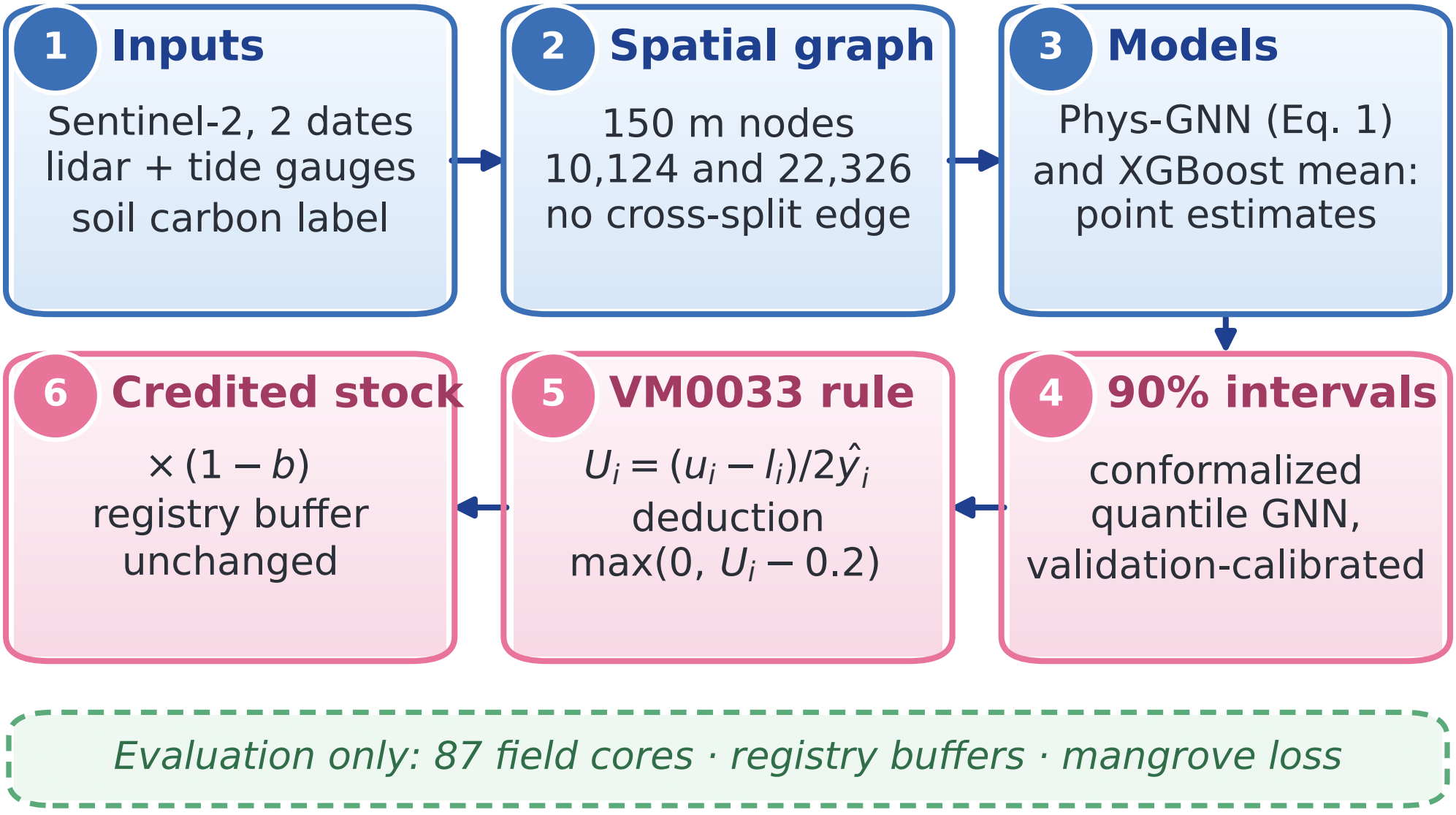

1 Inputs
Sentinel-2, 2 dates
lidar + tide gauges
soil carbon label
2 Spatial graph
150 m nodes
10,124 and 22,326
no cross-split edge
3 Models
Phys-GNN (Eq. 1)
and XGBoost mean:
point estimates
4 90% intervals
conformalized
quantile GNN,
validation-calibrated
5 VM0033 rule
$U_i = (u_i - l_i)/2\hat{y}_i$
deduction
max(0, $U_i - 0.2$)
6 Credited stock
$\times (1 - b)$
registry buffer
unchanged
Evaluation only: 87 field cores · registry buffers · mangrove loss